Trainings

General Information

Use this procedure **only** if the harness or connector can be repaired.

After performing any of the checks below, and it is necessary to repair or replace a harness or connector, refer to the table of contents in section 19 for the appropriate repair or replacement procedure.

Fault code troubleshooting trees will refer to this procedure when it is necessary to measure resistance on a harness, connector, or component that the fault code applies to. Each fault code troubleshooting tree will troubleshoot a particular component and the associated circuitry such as a pressure sensor, wiring harness and connectors that connect the sensor to the electronic control unit.

When troubleshooting to determine if a short or open exists in a particular circuit, all of the associated connectors, pins, circuit names and connections that apply to this component can be viewed on the applicable wiring diagram.

Use the following procedures to determine how to make the necessary resistance checks on components, connectors and circuits that apply to the fault code that referred you to this procedure.

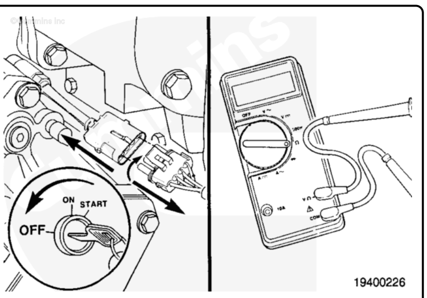
Resistance Check

Turn the key switch off.

Disconnect the appropriate connector from the component.

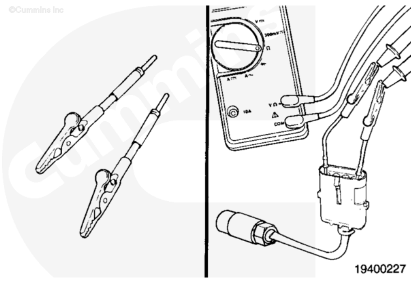
Adjust the multimeter to measure resistance.

Use the wiring diagram to determine the pins that apply to the component you are measuring.



⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the appropriate test lead for the connector. Refer to the Service Tools listing or the appropriate wiring repair kit.

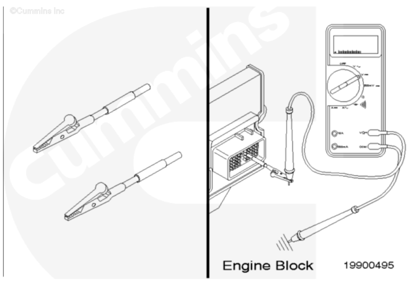


Connect the appropriate connector test leads to the connector pins and connect the alligator clips to the multimeter probe. Measure the resistance.

Compare this value to the applicable fault code specification or applicable Electrical or Sensor Specification on the wiring diagram. If the value is not correct, the component is malfunctioning. Refer to the applicable fault code procedure for instructions.

Continuity Check

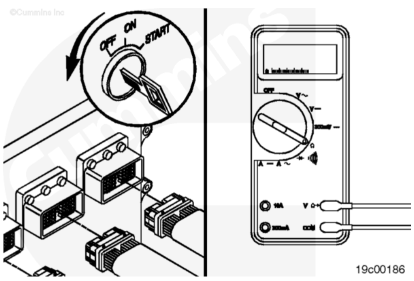
Continuity is an electrical connection between two pins that is less than a certain value. For harness wires, the specification is less than 10 ohms.



Turn the key switch to the OFF position.

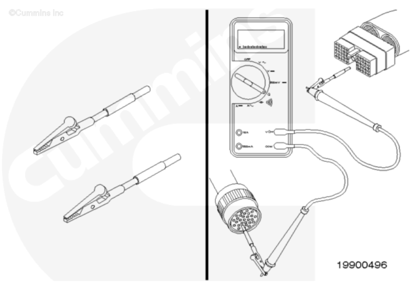
Disconnect the harness connectors that are to be tested.

Adjust the multimeter to measure resistance.



⚠ CAUTION ⚠

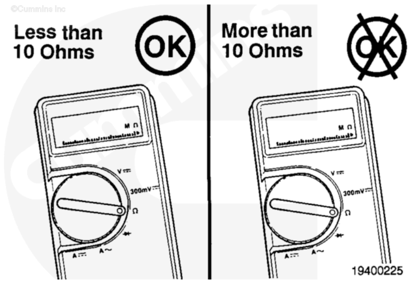
To reduce the possibility of pin and harness damage, use the appropriate test lead for the connector. Refer to the Service Tools listing or the appropriate wiring repair kit.



Connect the appropriate connector test leads to the connector pins and connect the alligator clips to the multimeter probe. Measure the resistance.

The multimeter **must** display less than 10 ohms for wire continuity. If the multimeter displays greater than 10 ohms, the wire **must** be repaired or the harness replaced.

Refer to the applicable fault code procedure for instructions.



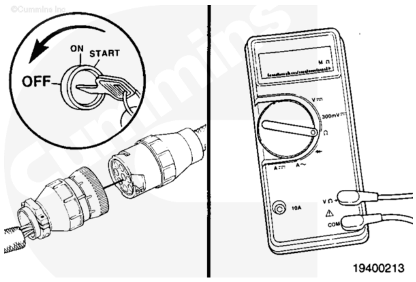
Check for Short Circuit from Pin to Pin

Short circuit from pin to pin check is a condition in which an electrical connection exists between two pins where it is **not** intended to exist.

Turn the key switch to the OFF position.

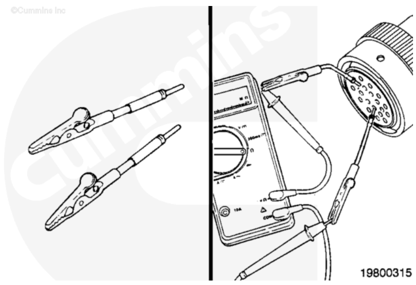
Disconnect the harness connectors that are to be tested.

Adjust the multimeter to measure resistance.



⚠ CAUTION ⚠

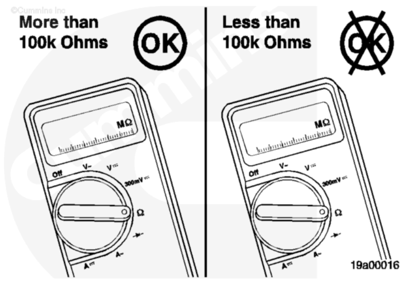
To reduce the possibility of pin and harness damage, use the appropriate test lead for the connector. Refer to the Service Tools listing or the appropriate wiring repair kit.



Connect the appropriate connector test leads to the connector pins and connect the alligator clips to the multimeter probes. Measure the resistance.

The multimeter **must** read greater than 100k ohms, which is an open circuit. If the circuit is **not** open, the pins being checked are electrically connected. Refer to the wiring diagram to verify that the wires are intended to be connected.

Inspect the harness connectors for moisture that can cause an inappropriate electrical connection.Refer to Procedure procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).



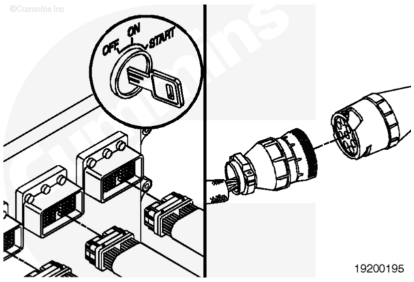
Refer to the applicable fault code procedure for instructions.

Check for Short Circuit to Ground

Short circuit to ground is a condition where a connection from a circuit to ground exists when it is not intended.

Turn the key switch to the OFF position.

Disconnect the harness connectors that are to be tested.



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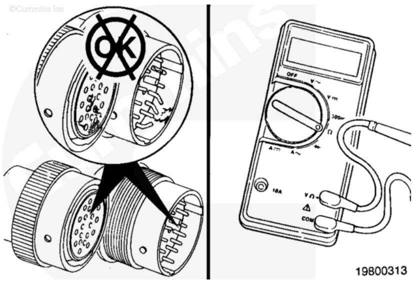
When testing a sensor, **only** the sensor connection is required to be disconnected.

When testing a harness, the harness connector at the electronic control unit and the connector at the sensor or multiple sensors **must** be disconnected.

Identify the pins that need to be tested.

Inspect the connector pins.Refer to Procedure procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).



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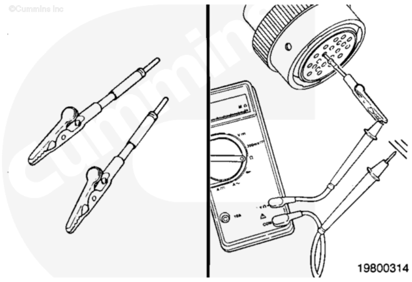
⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the appropriate test lead for the connector. Refer to the Service Tools listing or the appropriate wiring repair kit.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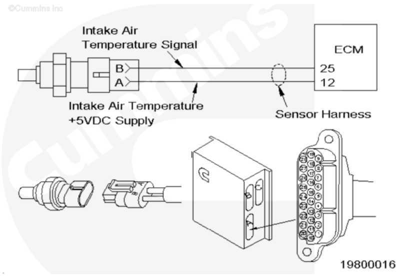
Connect the appropriate connector test lead to a connector pin and connect the alligator clip to the multimeter probe.

Touch the other multimeter probe to a clean, unpainted surface on the engine block or chassis ground. Measure the resistance.

The multimeter **must** read greater than 100k ohms, which indicates an open circuit. If the circuit is **not** open, the wire being checked has a short circuit to ground, the engine block or chassis ground.

Refer to the applicable fault code procedure for instructions.



Last Modified: 26-Mar-2012